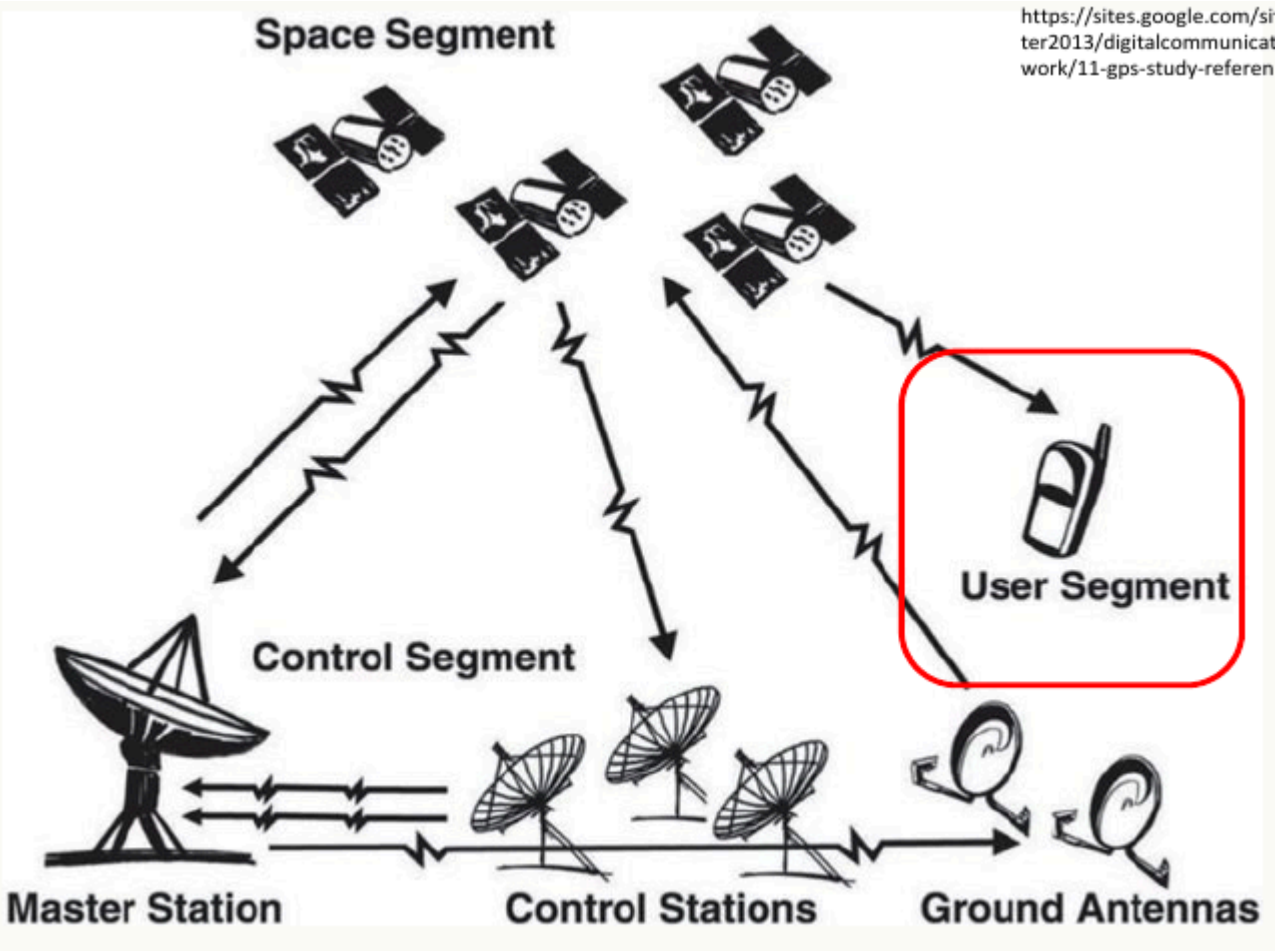
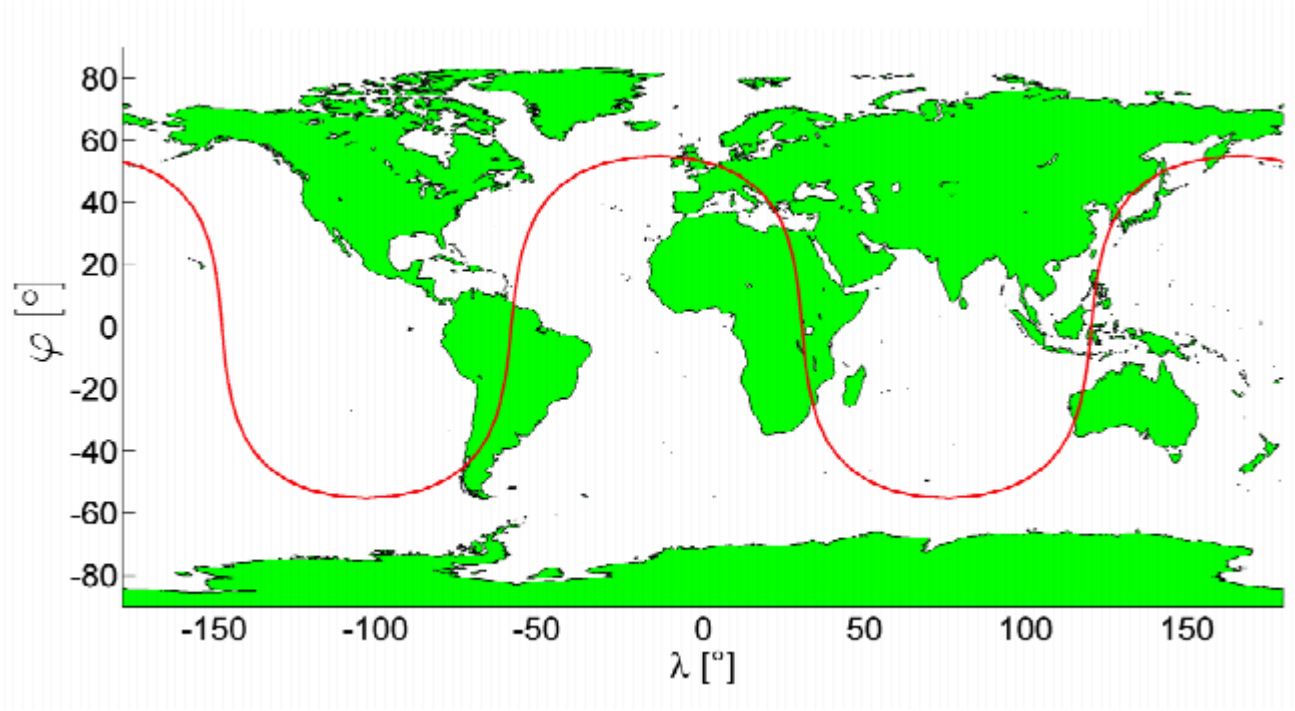
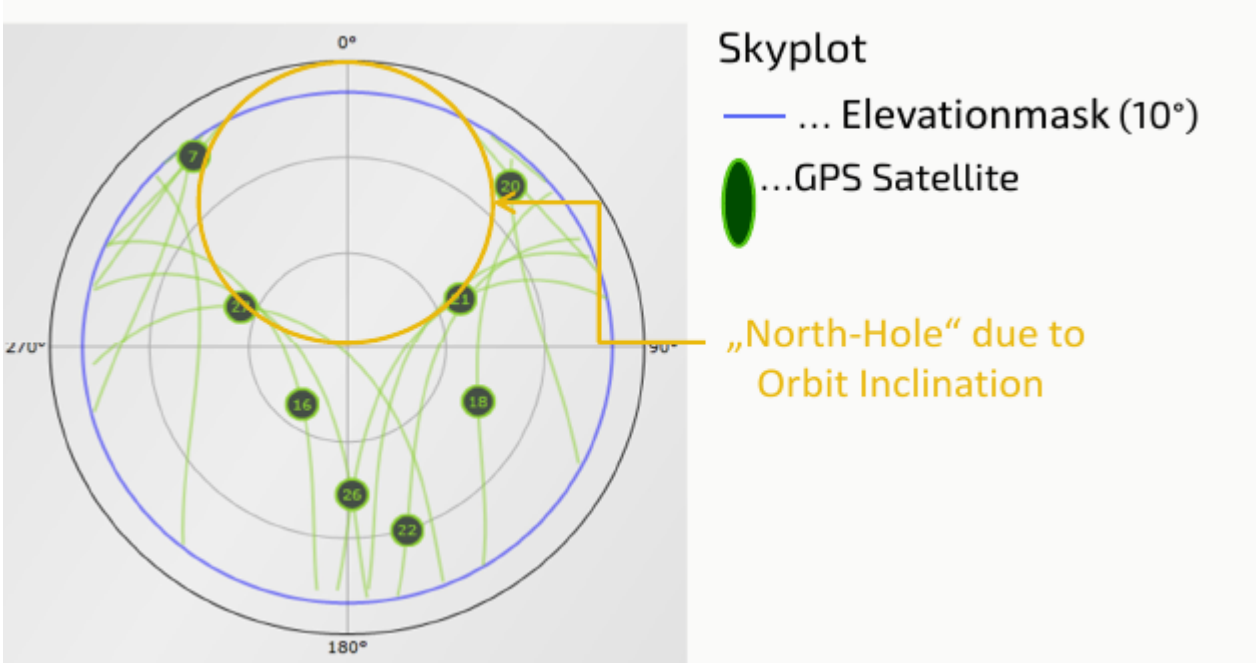
Lecture 2: GPS, Reference Systems & Satellite Orbits



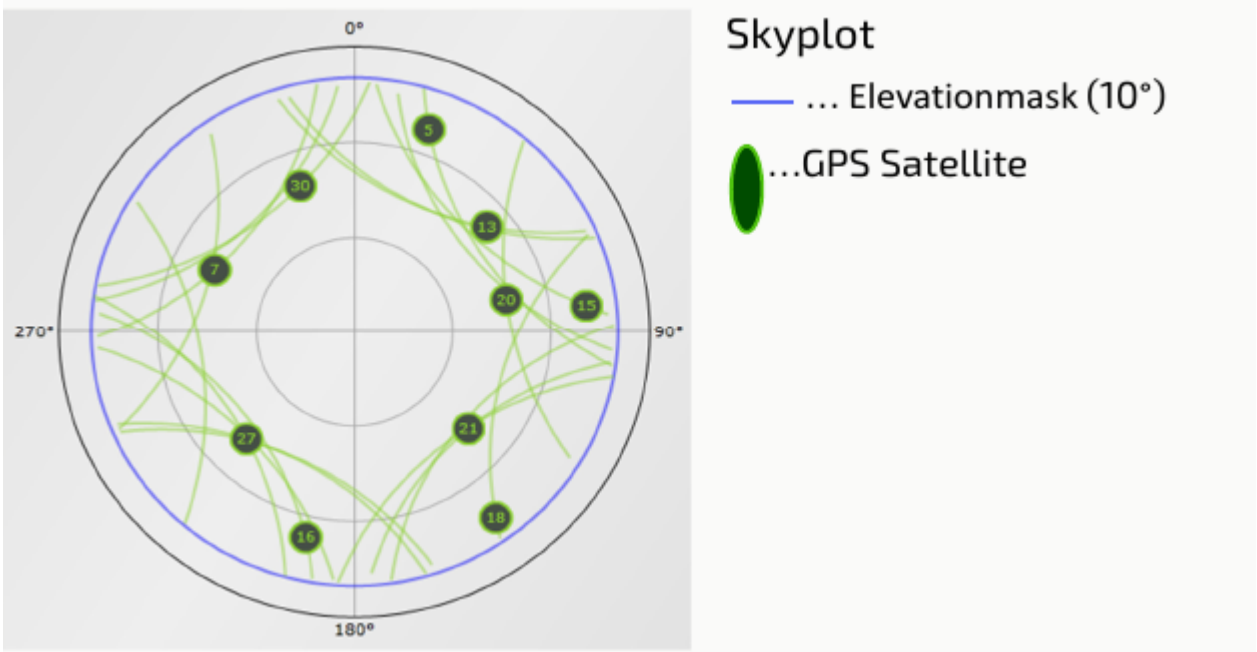
Ground Track of a GPS Satellite



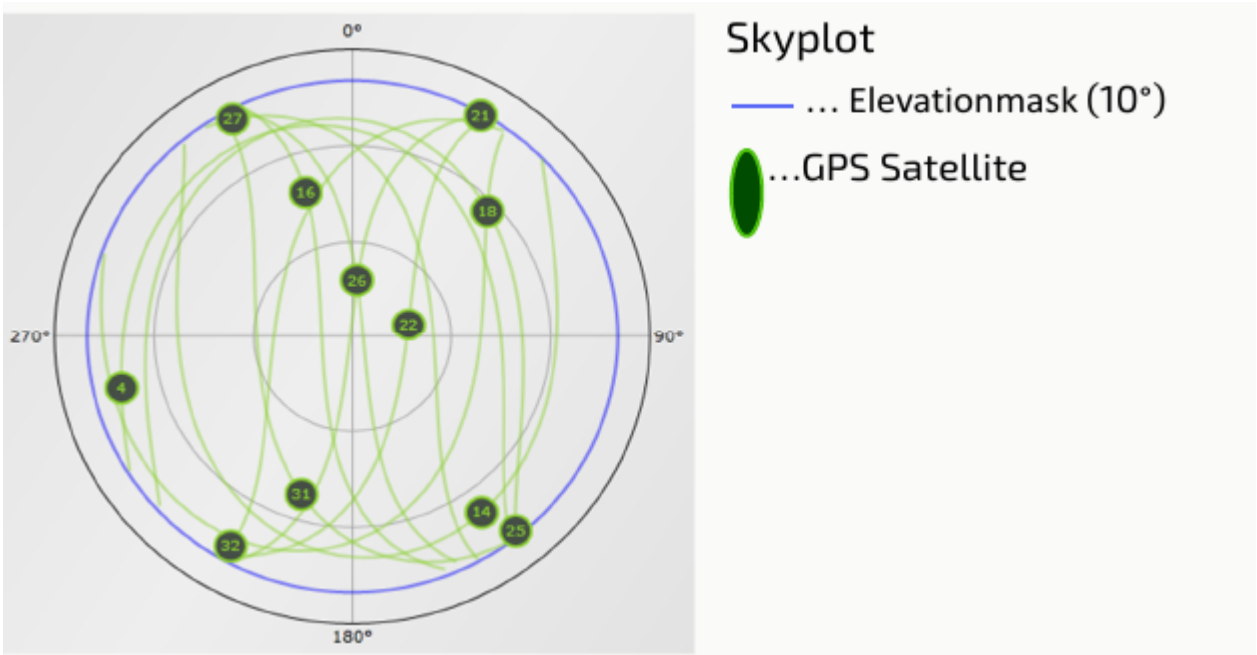
GPS Satellites in Bonn



GPS Satellites at Northpole



GPS Satellites at the Equator



Control Segment

- Control Stations located around the world
- Monitor and control satellite **orbits**
- Monitor and maintain satellite **health**
- Predict satellite orbits and satellite **clock** parameters
- Maintain GPS **time**
- Update **navigation message**

User Segment

- Satellite Receiver
  - Antenna
  - Receiver
- Processing Unit
- Optional:
  - Reference
  - Services

Navigation Message

- transmitted every 12.5 minutes
- Accurate info about transmitting satellites health, position and clock (**Broadcast Ephemerides**)
  - Repeats eery 30sec
- Not so accurate information about ALL Satellites health, position and clock (**Almanac**)
  - Repeats every 12.5 minutes

Almanac

Parameter	Explanation
ID	Satellite identification number
WEEK	Current GPS week
t_e	Reference epoch in seconds within the current week
$\sqrt{a}$	Square root of semimajor axis in $\sqrt{\text{meter}}$
e	Eccentricity
M_0	Mean anomaly at reference epoch
ω	Argument of perigee
δi	Inclination offset from 0.3 semicircles ($\hat{=}$ 54°)
ℓ_0	Longitude of the node at weekly epoch t_0
$\dot{\Omega}$	Drift of node's right ascension per second
a_0	Satellite clock offset in seconds
a_1	Satellite clock drift coefficient

- Clock Information
- Time Information
- Position Information (Kepler Elements, see later)

Broadcast Ephemerides

Parameter	Explanation
ID	Satellite identification number
WEEK	Current GPS week
t_e	Ephemerides reference epoch
$\sqrt{a}$	Square root of semimajor axis in $\sqrt{\text{meter}}$
e	Eccentricity
M_0	Mean anomaly at reference epoch
ω_0	Argument of perigee
i_0	Inclination
ℓ_0	Longitude of the node at weekly epoch t_0
Δn	Mean motion difference
$\dot{i}$	Rate of inclination angle
$\dot{\Omega}$	Rate of node's right ascension
C_{uc}, C_{us}	Correction coefficients (argument of perigee)
C_{rc}, C_{rs}	Correction coefficients (geocentric distance)
C_{ic}, C_{is}	Correction coefficients (inclination)
t_c	Satellite clock reference epoch
a_0	Satellite clock offset
a_1	Satellite clock drift coefficient
a_2	Satellite clock frequency drift coefficient

Need for reference systems!

Global Reference Systems

- Reference System
  - Definitions of a reference system using conventions, models and standards ("theoretical definition")
- Reference Frame
  - Practical realisation of a reference system using observations coordinates and fundamental reference points
  - Application of satellite and space geodesy methods (SLR, VLBI, GNSS)
  - Spatial coordinates are only valid when they are given with a time reference (Epoch)

CONVENTIONAL TERRESTRIAL REFERENCE **SYSTEM** (CTRS)

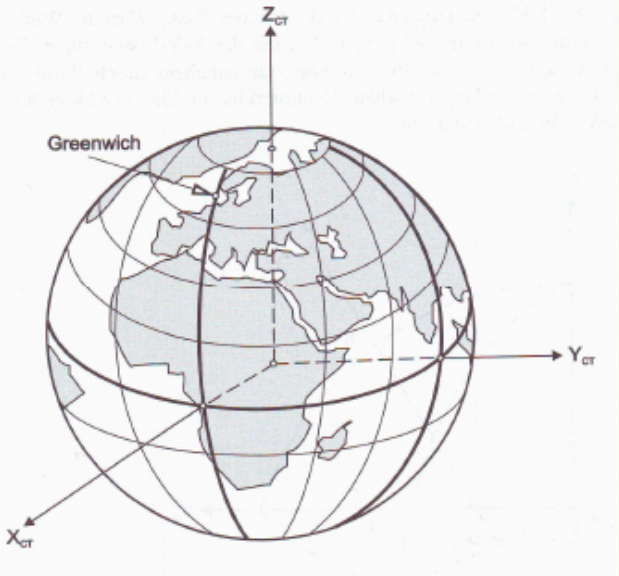


Fig.: Global geozentrisches Koordinatensystem
Source: Bauer, 2011

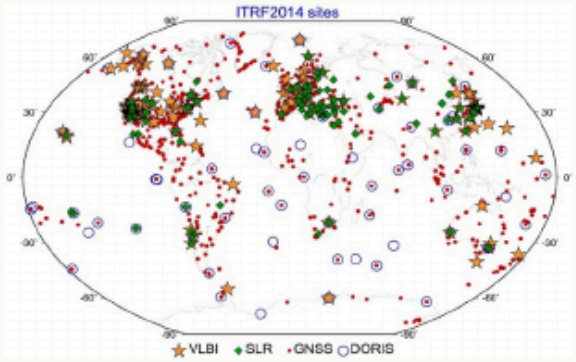
- Global gecentric system (ECEF, Earth Centred Earth Fixed)
- Used to describe the user position on Earth
- Origin: Center of Mass of the Earth
- Z-axis through rotation axis
- X-axis is intersection between Greenwich meridian and equatorial plane
- Y-axis completes right handed system

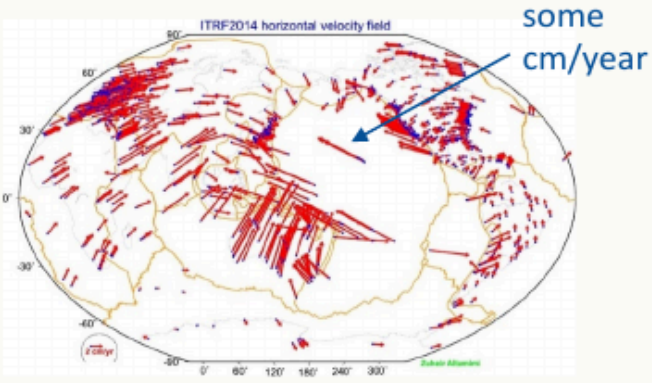
BUT: rotational axis is not constant (polar motion)

➔ z-Axis is going through the CTP (Conventional Terrestrial Pole, the average of the Earth pole between 1900 and 1905)

CONVENTIONAL TERRESTRIAL REFERENCE **FRAMES** (CTRF)

- Reference Frames are realizations of Reference Systems
- Because the axis and the origin are **not physically accessible** AND they **change in time**, the system is implemented by providing the coordinates at a number of dedicated points.
- Because of the motion of the Earth crust, the points have to be updated regularly and a velocity of the points is also given





- GPS Satellites are defined in the WGS 84
- GPS heights are related to the ellipsoid

Time Systems

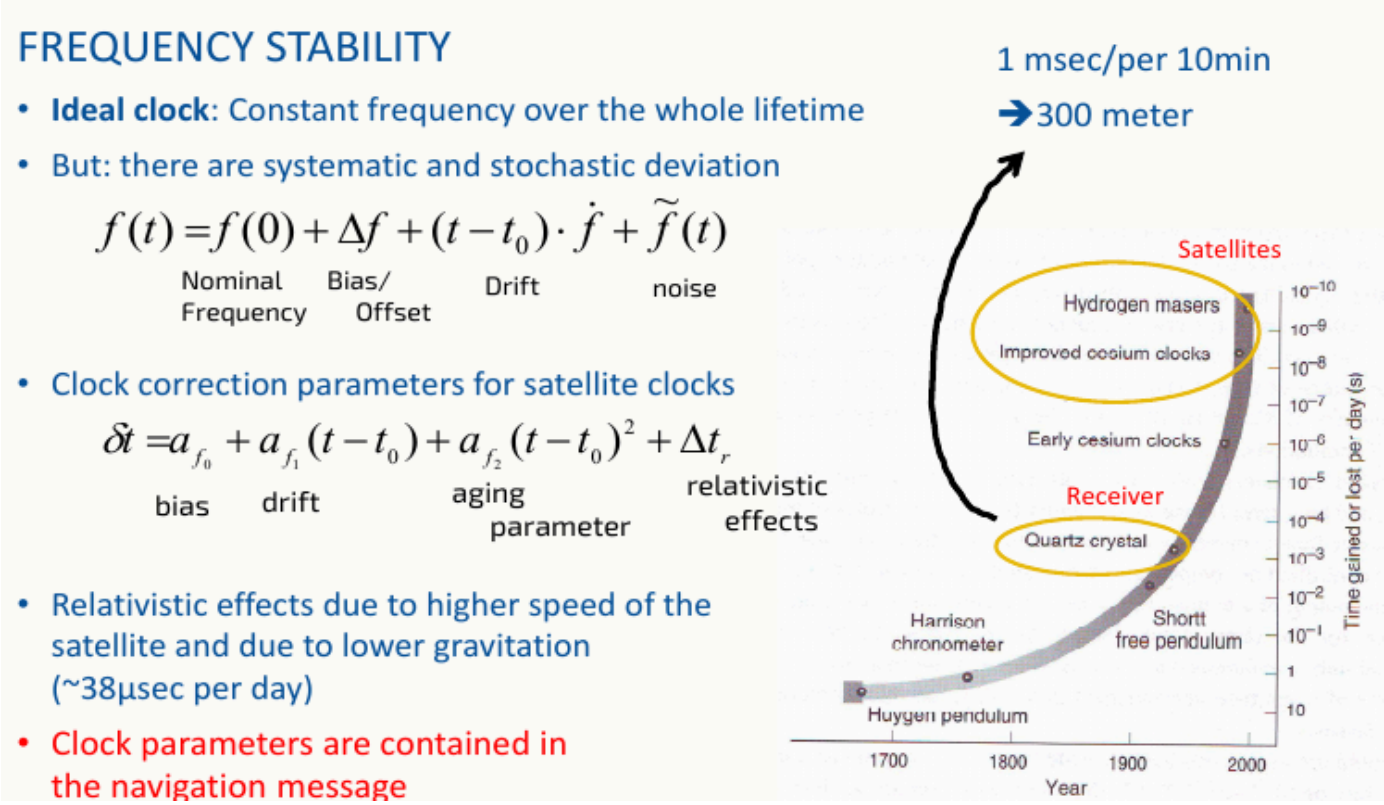
- **Universal Time (UT)**
  - defined by the actual rotation rate of the Earth
  - Not constant - Changes due to polar motions

    - Various corrected UT version exist (e.g. UT1, considering polar motions)
    - Scale of UT:
        - 1 solar day = 24 solar hours = 86400 solar seconds
- **International Atomic Time (TAI)**
    - Realised by over 200 high precision atomic clocks around the world
    - 1 SI day = 86400 SI seconds
    - Independant of Earth rotaton (= constant!)

-> UT1 and TAI are drifting away (Earth rotation slows down)

- **Coordinated UT (UTC)**
    - Scale: use Si seconds from TAI
    - Stay close to UT1 to "stay synced with Earth rotation"
    - Introduction of Leap Seconds to keep $|UT1 - UTC| < 0.9s$
    - Since January 2017: TAI = UTC +37 seconds
- **GPS time (GPST)**
    - Similar to UTC but without Leap seconds
    - based on atomic clocks at the GPS monitoring stations and satellites
    - Starttime = UTC(00:00:00, 6.1.1980), GPST -TAI = 19 seconds
    - UTC-GPST = 18 seconds @ January 2017 (leap seconds)
- **GPS week**
    - Start Sunday, 06.01.1980(00:00:00)
    - Max Value : 1023, 1st rollover Aug 99, April 2019, Nov 38
- **GPS seconds of Week (sow)**
    - Number of seconds from Sunday 00.00.00
    - 60480 seconds per week

Frequency stability

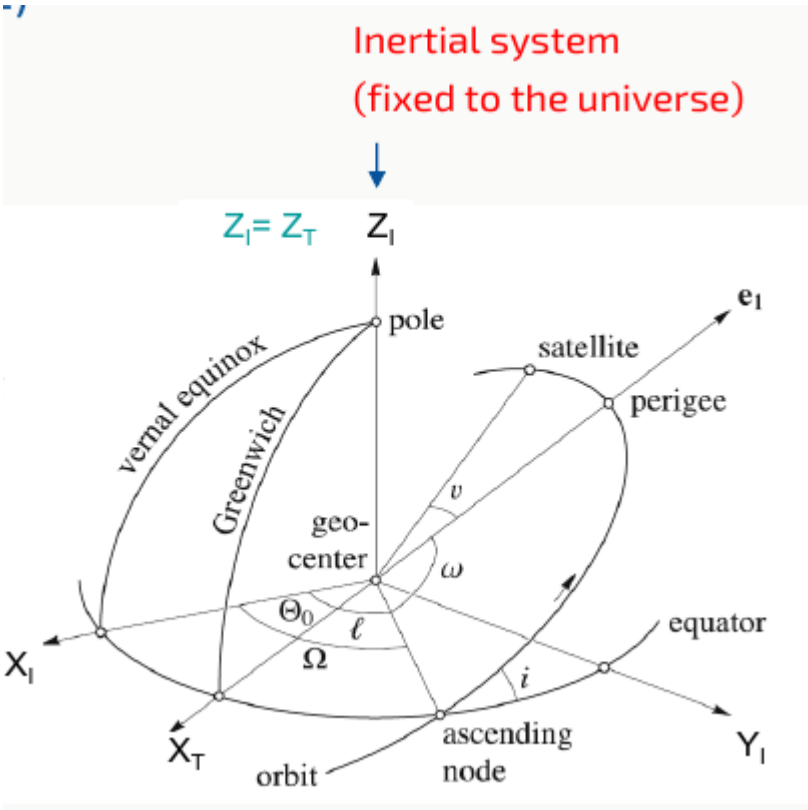


Kepler Elements (Ideal Case)

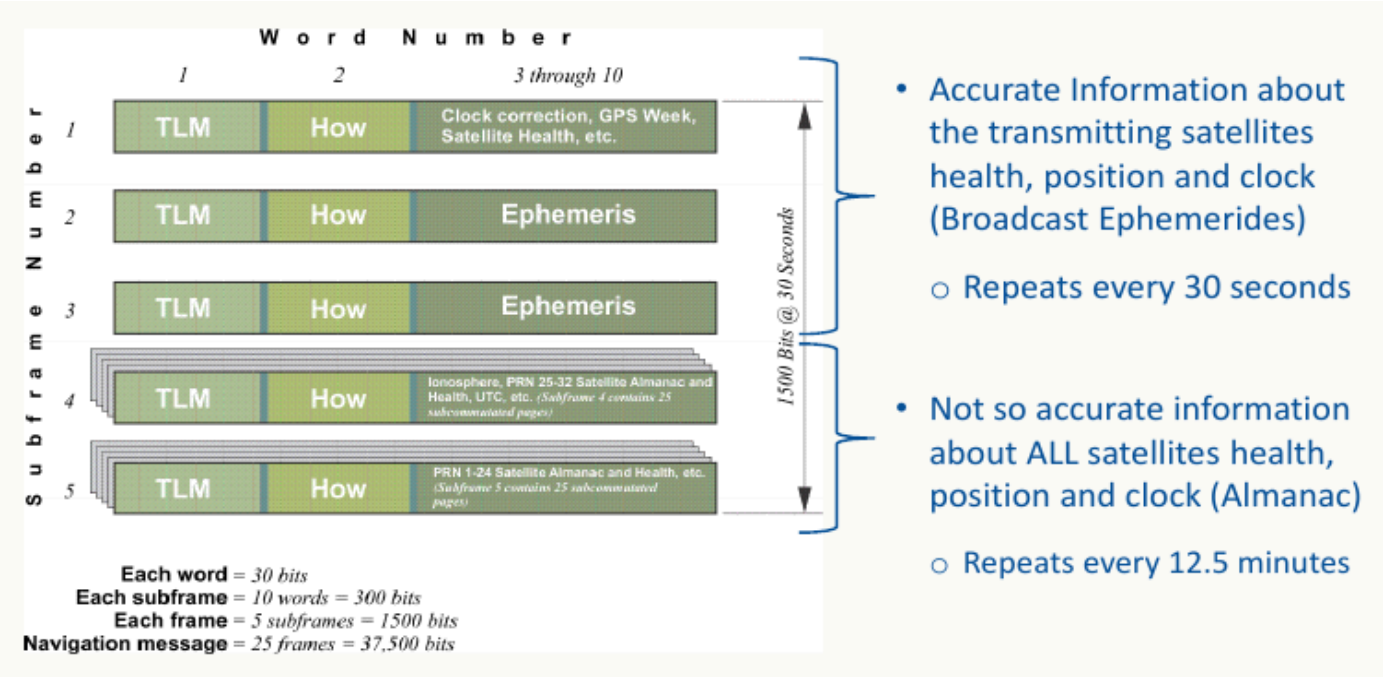
- Orbit size and shape
  - a semi-major axis
  - e eccentricity
- Orientation of the orbital plane
  - i Inclination
  - Ω Right ascension of th eascending node (RAAD)
- Orientation of the ellipse within the orbital plan
  - ω argument of pedigee
- Position within orbit
  - v True anomaly OR
  - tp epoch of perigee passage

Satellite Orbits

- Gravitaitonal Effects
  - Non-central gravitational field of the Earth
  - Gravitational fields of the sun and the moon
- Non gravitational Effects
  - Solar radiation pressure (if not in the earth shadow)
  - Atmospheric drag (collisions with atmospheric molecules)



Navigation Message



Almanac

- Accuracy: ~ km
- update: every 6 days
- Content (for all satellites)
  - General info
  - 6 Kepler - Elements +1 dift parameter

- 2 clock correction parameters

Parameter	Explanation
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WEEK	Current GPS week
t_a	Reference epoch in seconds within the current week
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e	Eccentricity
M_0	Mean anomaly at reference epoch
ω	Argument of perigee
δi	Inclination offset from 0.3 semicircles ($\hat{=}$ 54°)
ℓ_0	Longitude of the node at weekly epoch t_0
$\dot{\Omega}$	Drift of node's right ascension per second
a_0	Satellite clock offset in seconds
a_1	Satellite clock drift coefficient

Broadcast Ephemerides

- **Update Rate: 1-2 hours**
- Time valid: 4 hours (2 hours before and after reference epoch)
- **Accuracy 1-2m** (RMS)
- **Reference system: WGS 84**

Content:

1. General information
2. 6 Kepler elements (analogue to Almanac) +
   3 terms for correction of secular effects +
   3 terms for correction fo periodic effects
3. Correction parameters for satellite clock (2nd order polynom)

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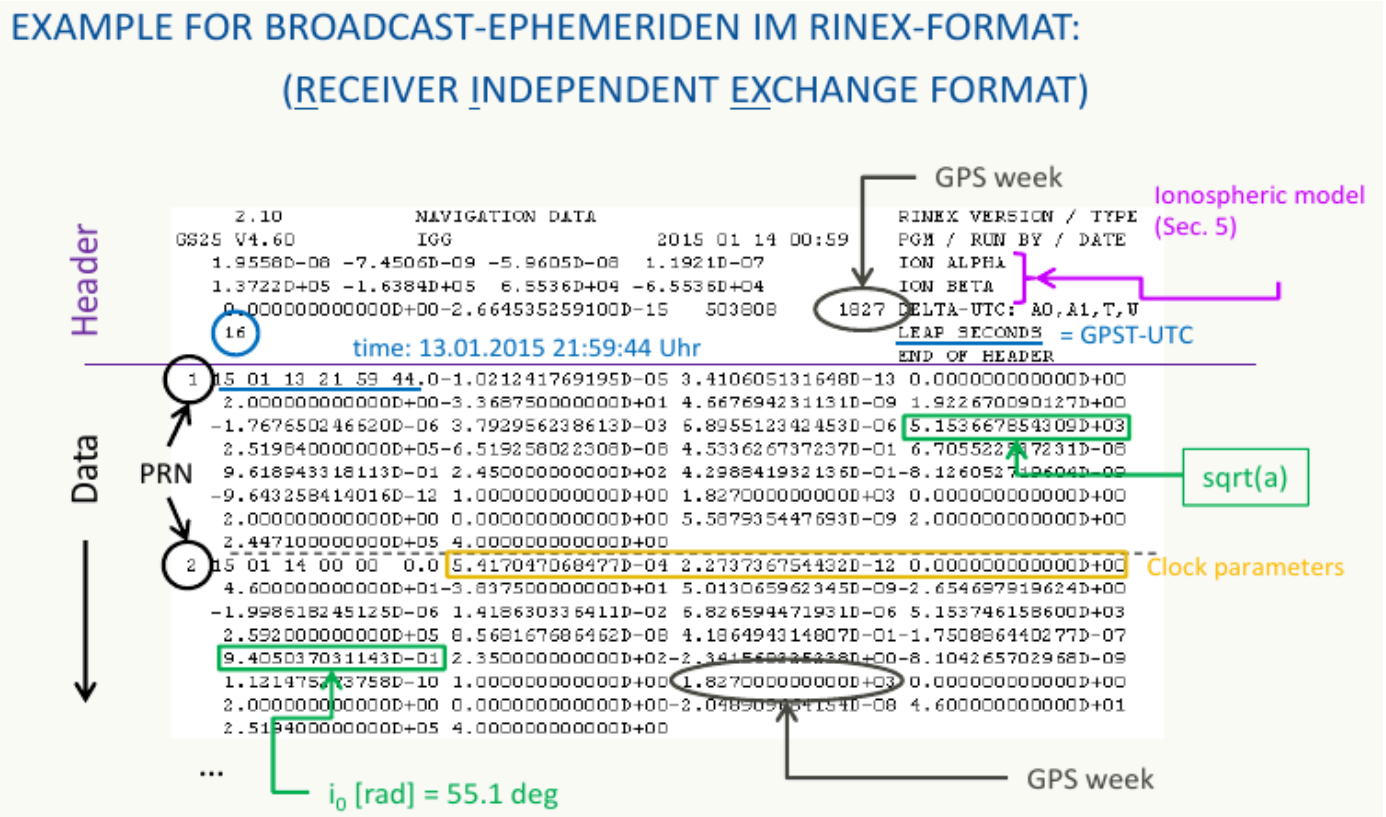

IGS Precise Orbits

- Founded in 1990by IAG (International Association of Geodesy)
- Main tasks:
  - Precise orbits (incl. clock parameters)
  - Current atmospheric parameters
  - Antenna calibration parameters
  - observations

- Scientific / civil service with, independant of the official (military controlled) results and also with higher accuracy
- Observations (carrier phase) from more than 400 globally distributed stations
- Orbit calculations at 10 different analysis centres
- Link to Terrestrial Reference Frames (TRFs) by connecting the GNSS station with other space geodetic stations (VLBI SLR)

IGS PRECISE ORBITS

DOWNLOADABLE FOR FREE FROM THE INTERNET

Type		Accuracy	Latency	Updates	Sample Interval
Broadcast	orbits	~100 cm	real time	--	daily
	Sat. clocks	~5 ns RMS ~2.5 ns SDev			
Ultra-Rapid (predicted half)	orbits	~5 cm	real time	at 03, 09, 15, 21 UTC	15 min
	Sat. clocks	~3 ns RMS ~1.5 ns SDev			
Ultra-Rapid (observed half)	orbits	~3 cm	3 - 9 hours	at 03, 09, 15, 21 UTC	15 min
	Sat. clocks	~150 ps RMS ~50 ps SDev			
Rapid	orbits	~2.5 cm	17 - 41 hours	at 17 UTC daily	15 min
	Sat. & Stn. clocks	~75 ps RMS ~25 ps SDev			5 min
Final	orbits	~2.5 cm	12 - 18 days	every Thursday	15 min
	Sat. & Stn. clocks	~75 ps RMS ~20 ps SDev			Sat.: 30s Stn.: 5 min

http://www.igs.org/products

Most accurate available orbits